

git - extra tools

Adapted from materials by Dr. Carrier



mergetools

There are external tools to help with merge conflicts

For example: meld

```
git config --global merge.tool meld
```

```
git config --global
```

```
mergetool.keepBackup false
```

After encountering a conflict with `git merge`, run
`git mergetool`

Still need to add and commit files afterward

Stashing changes

What if you have changes you want to *maybe* keep, but you don't want to commit? `git stash`!

`git stash` - saves both staged and unstaged changes and saves them for later

Add a `-u` to store untracked files, too!

`git stash pop` to then bring the changes back

There are more advanced features if you're interested!

Reverting changes

My group partner pushed an awful commit but you said not to rewrite history. What do I do?!?!?!?

```
git revert <commit>
```

Reverting changes

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```
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```

This does not rewrite history, it adds new history that reverses the mentioned commit :^)

(This is much better for syncing with other people)

Viewing previous states

I want to *look* at an old version of a file (but not bring back that old version):

```
git show <commit>:filepath
```

I want to bring back a previous version of a file:

```
git restore --source=<commit/branch>  
filepath
```

Old way:

```
git checkout <commit> filepath
```

Note: Careful! Checking out just a commit will detach HEAD!

Other GitHub features: forking

ferguson-gvsu / cis_241_w_2026

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<> Code

🕒 Issues

🔗 Pull requests

🔗 Agents

🕒 Actions

📁 Projects

📖 Wiki

More

cis_241_w_2026

Public

📌 Edit Pins

👁 Watch 0

🔗 Fork 0

★ Star 0

🔗 main

🔗

📄

Go to file

+

<> Code

FergusonAJ

Add the final lecture!

49140e5 · 6 months ago

🕒

📁 check_in_files	Add check-in file	6 months ago
📁 in_class_examples	Add make examples	6 months ago
📁 project_files	Add project files	9 months ago
📁 slides	Add the final lecture!	6 months ago
📄 README.md	Demonstrate cloning and...	8 months ago

About

⚙

Class materials for my Winter 2026 section of CIS 241 - System Level Programming and Utilities

📖 Readme

🔗 Activity

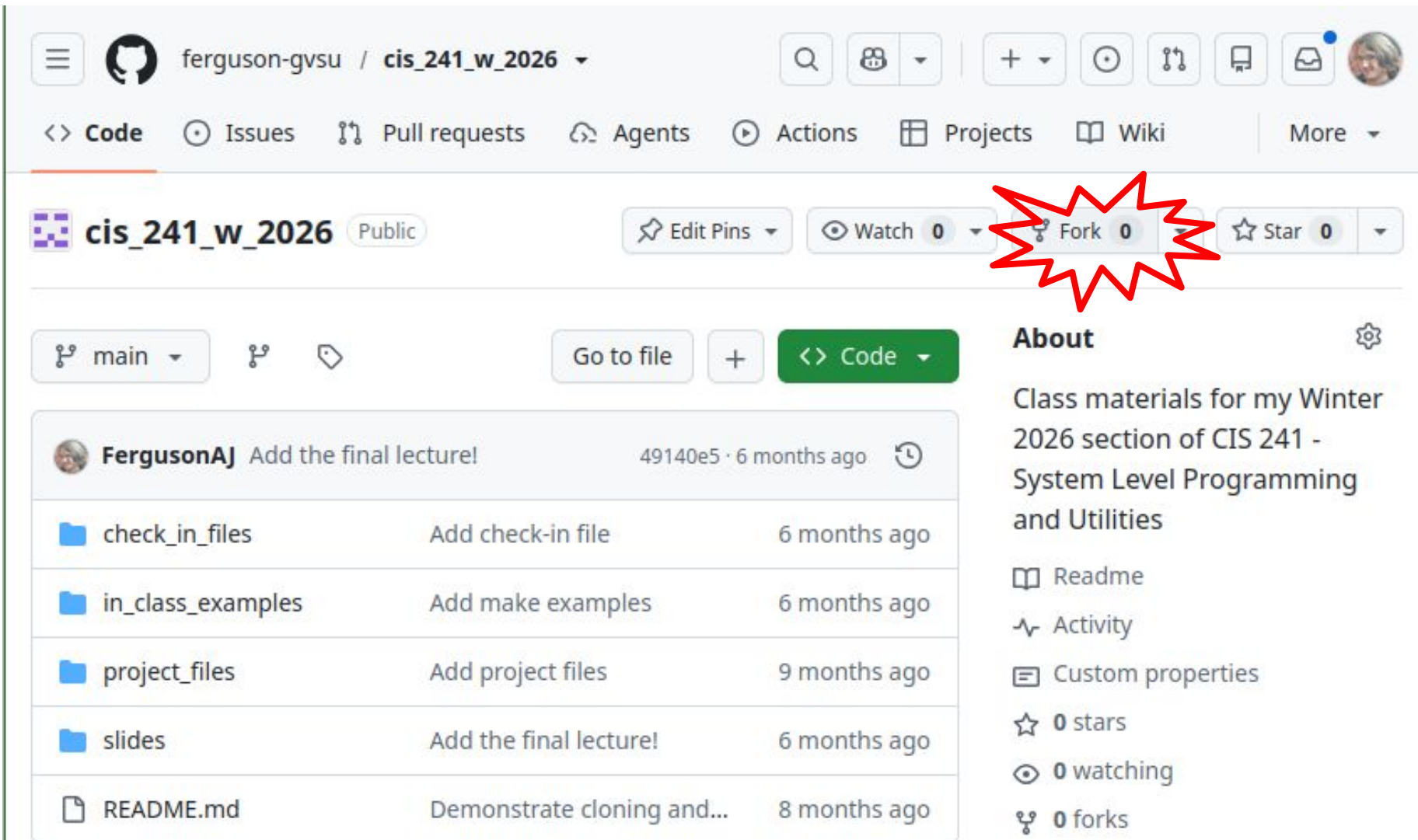
📖 Custom properties

★ 0 stars

👁 0 watching

🔗 0 forks

Other GitHub features: forking



The screenshot shows the GitHub interface for the repository 'ferguson-gvsu / cis_241_w_2026'. The repository is public and has 0 stars, 0 watches, and 0 forks. The 'Fork' button is highlighted with a red starburst. The file list shows folders for 'check_in_files', 'in_class_examples', 'project_files', and 'slides', along with a 'README.md' file. The 'About' section describes the repository as 'Class materials for my Winter 2026 section of CIS 241 - System Level Programming and Utilities'.

File/Folder	Description	Time
check_in_files	Add check-in file	6 months ago
in_class_examples	Add make examples	6 months ago
project_files	Add project files	9 months ago
slides	Add the final lecture!	6 months ago
README.md	Demonstrate cloning and...	8 months ago

Makes a copy of a repo and its history

- Typically only copies main branch, but can config
- You now have full control of that cop
- This is (often) how people contribute to others' repos!

Other GitHub features: pull requests

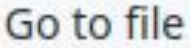
 FergusonAJ /  **MABE2** 



 **Code**  Pull requests  Agents  Actions  Projects  Wiki

 **MABE2** Public  Unpin  Watch **0** 

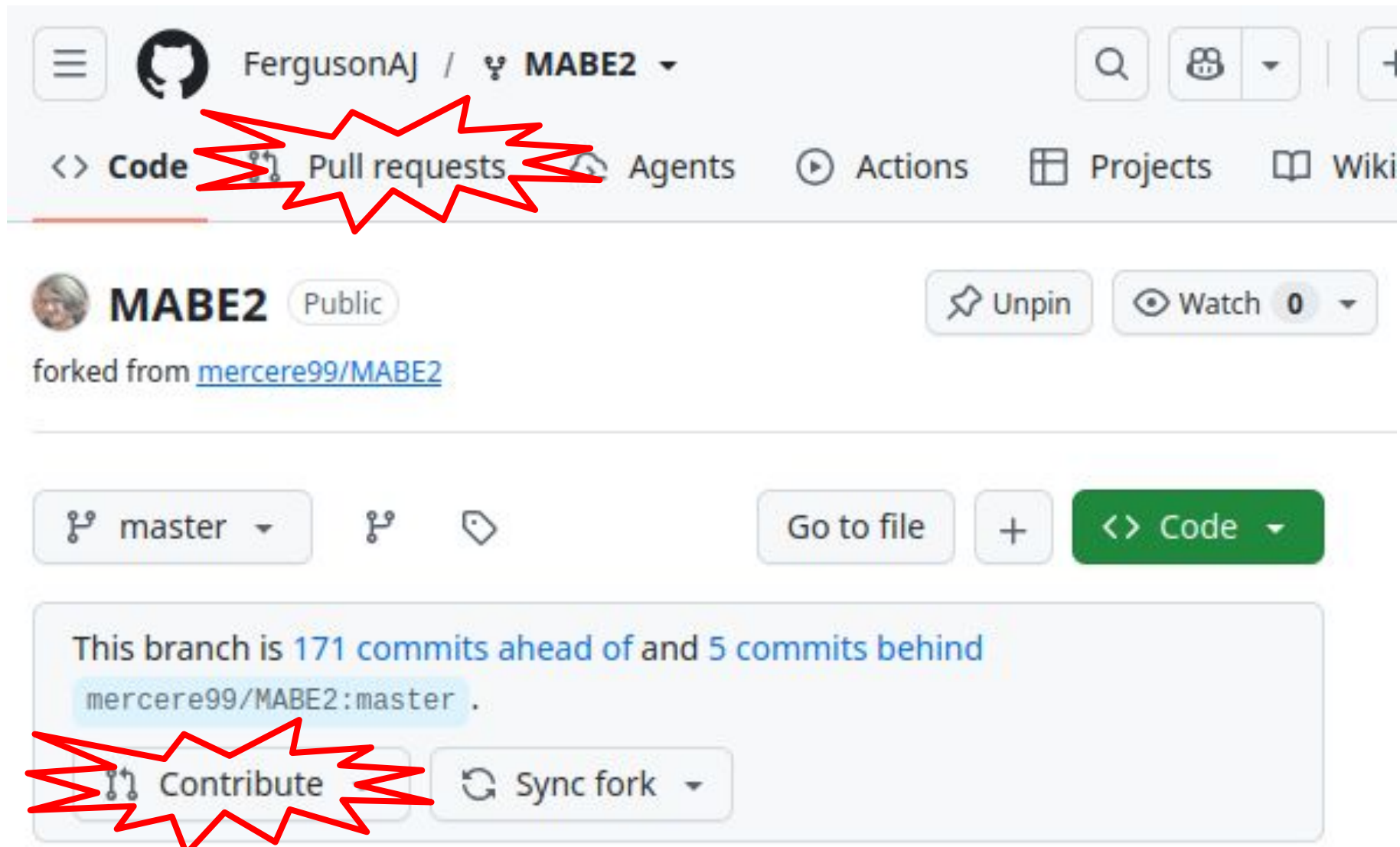
forked from [mercere99/MABE2](#)

 master       **Code** 

This branch is **171 commits ahead of** and **5 commits behind**
[mercere99/MABE2:master](#) .

 **Contribute**   **Sync fork** 

Other GitHub features: pull requests



GitHub-based merges [\(examples\)](#)

- Nice quality of life features (comments, [CI](#), etc)
- Can merge across forks

How the open-source ecosystem collaborates